MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE LS1SMWU (INCLUDING VARIATION LS1SMWA)

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the cable described herein shall consist of this document and the latest issue of MIL-C-24643.

This cable can be manufactured in two variations: LS1SMWU (unarmored) and LS1SMWA (armored).

REQUIREMENTS:

Qualification required.

Construction (watertight)

First - Copper conductor, in accordance with ASTM B 286, size 22-7, tin coated. Maximum diameter 0.033 inch.
Second - Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third - Cross-linked polyethylene insulation, white. Nominal thickness 0.030 inch. Standard identification code applied by method 1.
Fourth - Braided shield of AWG no. 36 tin coated copper. Braid angle of 30 to 35 degrees; minimum coverage of 85 percent.
Fifth - Shield insulation of two polyester tapes, type G in accordance with MIL-I-631, sealed.
Sixth - The required number of singles (see table I) cabled together with a lay not greater than 24 times the pitch diameter of the layer. Cabling sequence to be consecutive, starting with no. 1, from center outward. Fillers shall be employed to form a well-rounded cable.
Seventh - Binder tape applied helically with overlap.

FSC 6145

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Construction variant for type LS1SMWU

Eighth - Cross-linked polyolefin jacket. Jacket thickness, 0.050 inch, nominal. Cable surface marking required.

Construction variant for type LS1SMWA

Eighth - Cross-linked polyolefin jacket. Jacket thickness, 0.050 inch, nominal.
Ninth - Braided metal armor and paint.

TABLE 1. Details.

Military 1/ part no. M24643/47	Size 2/	Number of singles	Overall diameter (inches)			
			Type LS1SMWU		Type LS1SMWA	
			Minimum	Maximum	Minimum	Maximum
-01UN	70	70	1.465	1.555	1.515	1.605

1/ Unarmored. When option is required, see MIL-C-24643 for configuration of part number.
2/ Types LS1SMWU and LS1SMWA.

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - ohms/1000 feet, at 25°C, maximum.	17.71
Voltage withstand - volts, root mean square, minimum	
Conductor to shield.....	5000
Shield to shield.....	500
Insulation resistance - megohms/1000 feet, minimum.....	500
Conductor continuity.....	No failure
Group A:	
Visual and dimensional.....	No failure
Watertightness - see MIL-C-24643 for limits of water leakage.....	No failure
Capacitance - pF/ft at 1 kHz maximum.....	35
Attenuation - at 400 MHz, decibel per 100 feet, maximum.	18
Group B:	
Cold bending, cable - at 0 ± 2°C, mandrel maximum diameter 20 inches, maximum.....	No damage
Drip - 95 ± 1°C.....	Zero
Tear - pounds per inch thickness, minimum (ASTM D 470)..	35
Physicals (unaged)	
Insulation	
Tensile strength - lb/in^2, minimum.....	1800
Elongation - percent, minimum.....	250

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EXAMINATION AND TESTS: (Continued)

	Requirements
Jacket (cable)	
Tensile strength - lb/in^2, minimum.....	1300
Elongation - percent, minimum.....	160
Group C:	
Physicals (aged)	
Insulation	
Air oven	
Tensile strength - percent of unaged, minimum.....	80
Elongation - percent of unaged, minimum..	80
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum.....	60
Elongation - percent of unaged, minimum..	60
Hot oil immersion	
Tensile strength - lb/in^2, minimum.....	50
Elongation - percent, minimum.....	50
Shrinkage.....	No failure
Permanence of printing (conductor) - cycles, minimum....	50
Permanence of printing (jacket) type LSISMWU - cycles, minimum.....	250
Cable filler removability.....	No failure
Heat distortion - percent of unaged, maximum.....	30
Armor - conformance to material, construction and coverage, type LSISMWA only.....	No failure
Shield - conformance to material, construction and coverage.....	No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B and C, plus the following:

Aging and compatibility (cable) (125 ± 5°C).....	No failure
Abrasion resistance (jacket/insulation).....	No failure
Acid gas equivalent - percent, maximum	
Jacket.....	2
Fillers.....	2
Insulation.....	18
Halogen content - percent, maximum	
Jacket.....	0.2
Fillers.....	0.2
Flame propagation (cable).....	No failure
Immersion (jacket)	
Tensile strength - percent of unaged, minimum.....	50
Elongation - percent of unaged, minimum.....	50

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QUALIFICATION INSPECTION:

	Requirements
Smoke index, maximum	
Jacket	25
Fillers	--
Insulation	10
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTH: 1000 feet (nominal)

NOTE: Not for Air Force use.

Custodians:
Army - MI
Navy - SH

Preparing activity:
Navy - SH
(Project 6145-0964)

Review activities:
Army - AV, CR, ER
Navy - EC
DLA - IS

User activities:
Army - ME, AR, AL
Navy - CG

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